Logical volume management (LVM):

- Volumes can consist of more than one disk.
- Easy resize operation.
- Easy replacement of failing disks.
- Advanced options such a working with snapshots, which allows you to create backups even if they are open.
- Easy to add new volumes.
- Easy to add many volumes.
- Up to 256 logical volume.

steps:

1- Partition physical storage

2- Create physical volume (PV)   (LVM automatically segments PVs into physical extents (PE))

3- Create volume group(VG)   (PV can only be allocated to a single VG)

4- Create logical volume (LV)

- Mirroring causes each Logical Extent to map to two Physical Extents.

[root@master ~]# pvcreate /dev/sdb1 /dev/sdc1 /dev/sdd1   (label the partition for use with LVM)

[root@master ~]# pvdisplay

[root@master ~]# pvdisplay /dev/sdb1

[root@master ~]# pvs

[root@master ~]# vgcreate VG1 /dev/sdb /dev/sdc1 /dev/sdd1

[root@master ~]# vgdisplay

[root@master ~]# vgdisplay VG1

[root@master ~]# vgs

[root@master ~]# lvcreate -n LV1 -L 2G VG1

[root@master ~]# lvdisplay

[root@master ~]# lvdisplay /dev/VG1/LV1

[root@master ~]# lvs

[root@master ~]# mkfs.xfs /dev/VG1/LV1

[root@master ~]# mkdir data

[root@master ~]# mount /dev/VG1/LV1 data

[root@master ~]# df -h

- Removing a logical volume will destroy any data stored on the logical volume.

[root@master ~]# lvremove /dev/VG1/LV1   (file system must be unmounted first)

[root@master ~]# vgremove VG1

[root@master ~]# pvremove /dev/sdb1 /dev/sdc1 /dev/sdd1

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Extending Logical Volumes (no down time):

[root@master ~]# pvcreate /dev/sde1

[root@master ~]# vgextend VG1 /dev/sde1

[root@master ~]# lvextend -L +3G /dev/VG1/LV1

[root@master ~]# xfs_growfs /dev/VG1/LV1   (update the file system for XFS file systems)

[root@master ~]# resize2fs /dev/VG1/LV1   (update the file system for other file systems)

Or:

[root@master ~]# lvextend -r -L +3G /dev/VG1/LV1   (extend and update in one step)

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Shrinking a volume group:

- XFS doesn't support shrinking.

[root@master ~]# umount data

[root@master ~]# resize2fs /dev/VG1/LV1 100M

[root@master ~]# e2fsck -f /dev/VG1/LV1

[root@master ~]# lvreduce --size -3G /dev/VG1/LV1

[root@master ~]# lvreduce --size -r -3G /dev/VG1/LV1

[root@master ~]# vgreduce VG1 /dev/sde1   (removes sde1 from VG1)

[root@master ~]# mount /dev/VG1/LV1 data

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Device mapper:

- The kernel uses the mapper to connect to storage devices such as LVM, RAID, LUCKS.

[root@master ~]# ll /dev/dm-0

[root@master ~]# ll /dev/mapper/VG1-LV1

[root@master ~]# ll /dev/VG1/LV1